

## URL shortner

### Functional requirements:

user should be able to shorten original URL ,  
shortened URL should be unique,  
end-user should be redirected to appropriate page as per his preference,

### NOn-functional requirements:

Data-partitioning,  
load balancing,  
fault tolerance,  
vertical scalability,  
horizontal scalability,  
low latency,  
high throughput,  
CAP theorem (Here Eventual consistency is fine , i.e 99.99% uptime . i.e AVailability>>Consistency)

### Core Entiites:

user ,  
url,  
history,

### API :

POST:/shorten

```
{  
  userID,  
  originalURL  
}
```

GET:/click/{url}

### HLD :

